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Original Research Article

Antimicrobial resistant *Shigella* in North India since the turn of the 21st century

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A R T I C L E I N F O

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A B S T R A C T

Purpose: The ubiquitous presence and rampant spread of antibiotic resistant strains of *Shigella* spp is a major public health concern. Therefore, monitoring the trends of antimicrobial resistance in them is essential.
Methods: A total of 15440 stool samples were inoculated on MacConkey agar, lysine deoxycholate agar and Selenite F enrichment broth from 2001 to 2015.Out of 491 shigellae isolated, 250 isolates were recovered from our culture collection. Antimicrobial susceptibility was performed by Kirby Bauer disc diffusion method, E-test and phenotypic resistance screening for ESBL and AmpC production was performed. For the detection of beta-lactamase genes, PCR for *bla*TEM, *bla*SHV, *bla*OXA, *bla*CTX-M-15, CMY-2 and *mph*A PCR in isolates with decreased susceptibility to azithromycin(DSA) was performed.
Results: S. flexneri (n = 173) was most common, followed by *S.dysenteriae* (n = 24), *S.sonnei* (n = 23), *S.boydii* (n = 10) and Non agglutinating *Shigella* (NAG, n = 20). A see-saw pattern in the prevalence of *S. flexneri* and *S. dysenteriae* and rising prevalence of *S. sonnei* and NAG was seen. Majority (77%) of the isolates had MICs >4 mg/L for ciprofloxacin and >50% had high MIC_{90} (12 mg/L) for ceftriaxone and cefepime (8 mg/L). Nearly 20% of *S.flexneri* were resistant to third generation cephalosporin by disc diffusion and 33.7% had MIC ≥ 1 μg/mL. Among the ceftriaxone resistant isolates (n = 29) the *bla*TEM beta-lactamase resistance gene was seen in all, *bla*CTX-M-15 in 37%, *bla*CMY-2 in 45.6% and *bla*OXA in 52%. The first report of DSA at our institute was in 2001 (n = 1, 2.5%) which increased to 35.1% (n = 40) in 2011–15. The isolates with DSA included *S. flexneri* (n = 40), *S. boydii* (n = 4) and *S. sonnei* (n = 1) and plasmid mediated resistance to azithromycin by *mph*A gene was detected in 19 out of 40 isolates of *S. flexneri.*
Conclusion: Global emergence of resistance *Shigella* is a matter of concern and calls for systematic monitoring and periodic updates of countrywide and local antibiogram.

1. Introduction

Shigella species are reported to cause acute gastroenteritis in a large number of children worldwide, with the most devastating consequences seen in the developing world [1]. Annually, an estimated 164.7 million cases of shigellosis are seen globally, of which the majority occur in developing nations, contributing to nearly 1.1 million deaths with 61% of all deaths involving children under the age of 5 years [2]. Apart from children, shigellosis also affects groups such as persons in custodial organizations, homeless persons and men having sex with men (MSM), and can spread quickly in these groups resulting in focal outbreaks [3]. The World Health Organization (WHO) recommends that all cases of bloody diarrhea should be treated promptly with an antimicrobial that is known to be effective against *Shigella* to reduce the risk of complications, duration of sickness and mortality and also reduce the risk of transmission to susceptible contacts [4].

The ubiquitous presence and rampant spread of shigellosis in Asian and African countries with the emergence of antibiotic resistant strains is particularly concerning and has counteracted the control attempts of public health communities [5]. Multidrug resistant (MDR) *Shigella* are a

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therapeutic challenge across the world, and in India with reports of resistance to third generation cephalosporins, fluoroquinolones, and even azithromycin [1].

Ours is a tertiary care, referral hospital in the city of Chandigarh at North India, catering to 7 adjoining states. Continuous monitoring of antibiotic resistance in *Shigella* is being carried out here and in this study, we present the data of antimicrobial resistance of shigellae isolated from our center over a period of 15 years. We also studied the genetic determinants of resistance in *Shigella* isolates in our culture collections. In addition, a comparison of the present findings with previously published data from 1990 to 2000 is also presented herein.

2. Material and methods

2.1. Sample collection

The geographic locations of the patient residence among the cases whose stool samples were collected in different parts of North India are shown in Fig. 1. This included seven states viz. Chandigarh, Punjab, Haryana, Uttar Pradesh, Himachal Pradesh, Rajasthan and Jammu and Kashmir. The samples were collected from patients suffering with diarrhea/dysentery from 2001 to 2015, in Cary Blair medium and transported to the Enteric laboratory at PGIMER. The study was approved by the Institute Ethics Committee and received partial funding from the Indian Council of Medical Research (ICMR).

2.2. Bacterial strains

The stool specimens were inoculated in MacConkey agar (MCA) and lysine deoxycholate agar (XLD) (Himedia Laboratories Pvt. Ltd, Mumbai, India) and Selenite F enrichment broth which was sub-cultured onto MCA and XLD after 6 h. Identification of non-lactose-fermenting colonies was performed using a biochemical test panel. Confirmatory identification was carried out by slide agglutination using polyvalent antisera (Denka-Seiken, Tokyo, Japan). The isolates which did not agglutinate on serotyping were reported as non-agglutinable (NAG) *Shigella*. All isolates of *S. flexneri, S. dysenteriae, S. sonnei* and *S. boydii* isolated from 2000 through 2015 were stored in brain-heart infusion (BHI) broth with 15% glycerol at −70 °C. These isolates were revived from our culture collection for the purpose of this study. We also retrospectively recorded the species distribution and antimicrobial susceptibility profile of isolates recovered from 1990 to 1999 for comparison with present findings.

2.3. Antimicrobial susceptibility testing

Kirby Bauer disc diffusion method was used to determine the antimicrobial susceptibility. The following antibiotics (Himedia Laboratories Pvt. Ltd, Mumbai, India) were used: ampicillin (AMP,10 μg), cefotaxime (CTX, 30 μg), cefepime (CPM, 30 μg), ceftriaxone (CTR, 30 μg), nalidixic acid (NA, 30 μg), ciprofloxacin (CIP, 5 μg), norfloxacin (NX, 10 μg), chloramphenicol (C, 30 μg), co-trimoxazole (COT, 25 μg) and azithromycin (AZM, 15 mg) and the interpretation was carried on as per the CLSI guidelines [6]. *Escherichia coli* 25922 ATCC strain was used as control during antimicrobial susceptibility testing.

The MIC determination was performed for cefotaxime, ceftriaxone, cefepime, ciprofloxacin and azithromycin by an E-test (bioMe'rieux India Pvt. Ltd, New Delhi, India). MIC breakpoints were considered as per the CLSI guidelines [6]. The MIC of >8 μg/mL was defined as decreased susceptibility to azithromycin (DSA) for *S. flexneri* and >16μg/ml for *S. sonnei*. Phenotypic screening for ESBL production was performed for all isolates showing higher MIC for ceftriaxone and cefepime by disc approximation method as per CLSI guidelines. All isolates with cephalosporin resistance were screened for AmpC by the EDTA disc method [7]. The PCR for *bla*TEM, *bla*SHV, *bla*OXA, *bla*CTX-M-15 and CMY-2 was

Fig. 1. Areas of stool sample collection from patients with diarrhea/dysentery from 2001 to 2009, in North India.

performed. For the isolates with DSA, the PCR for detection of the *mph*A gene was also performed. The details of all primers used is provided in Table 1.

2.4. Statistical analysis

The data was analyzed using the SPSS software (version 20.0) and prism graph pad (version 6.0). The results were tabulated in addition to our previous data

3. Results

3.1. Isolation of Shigella species

A total of 15440 stool samples were received and 491 (3.18%) isolates of *Shigella* spp were recovered from 2001 to 2015 of which 250 isolates could be revived. Phenotypic identification revealed that the majority were *S. flexneri* (n = 173) followed by *S. dysenteriae* (n = 24), *S. sonnei* (n = 23), *S. boydii* (n = 10) and NAG (n = 20). The species distribution was studied over equal 5-year intervals, 2000–05 (n = 43), 2006–10 (n = 95) and 2010–2015 (n = 112). The distribution of *Shigella* species from 2000 to 2015 (15 years) along with retrospectively collected data from 1990 to 1994 (n = 62) and from 1995 to 1999 (n = 52) is shown in Fig. 2. The prevalence of *S. flexneri* was accompanied by an inverse trend in the prevalence of *S. dysenteriae*. Prevalence of *S. sonnei* ranging from 3.2% in 1990–1994 to 8% in 2011–2015 and from 4.8% in 1990–1994 to 2.7% in 2011–1015 for *S. boydii* was noted. A rise in the NAG was seen which emerged in 2006–2010 (3.1%) showing a nearly 5 times rise to 15.2% in 2011–2015.

3.2. Antimicrobial susceptibility results

A comparison of the antimicrobial susceptibility findings between 2000 and 2015 is depicted in Fig. 3. The first report of ciprofloxacin resistance was in the year 2000 in a single isolate of *S. flexneri.* The overall trends of proportion of antimicrobial resistance seen from 1990 to 2015 is shown in Fig. 4.

The results of MIC testing revealed that 77% isolates had clinically ineffectual MICs for ciprofloxacin (>4 μg/mL) and more than 50% had very high MIC to ceftriaxone. The MIC 50 and MIC90 for ceftriaxone were 0.125 mg/L and 12 mg/L respectively. The MIC_{90} for cefepime was 8 mg/L. Both the MIC 50 and MIC 90 for azithromycin were ≥256 μg/mL.

Till 2009, all isolates of *S. sonnei* were susceptible to ciprofloxacin. However, in 2015, more than 75% of them were reported to be resistant with a MICs of >32 μg/ml. A total of 5 ciprofloxacin resistant *S. boydii* isolates were seen (one in 2005 and 4 in 2015). Nearly 20% of *S.flexneri* were resistant to third generation cephalosporin by disc diffusion and 33.7% exhibited an MIC≥1 μg/mL. Among the ceftriaxone resistant isolates (n = 29) the detection of beta-lactamase resistance genes revealed 100% isolates to be positive for *bla*TEM, 37% for *bla*CTX-M-15, 45.6% for *bla*CMY-2 and 52% for *bla*OXA; none of the isolates was positive for *bla*SHV.

Overall, azithromycin resistance (MIC > 8 μg/mL for *S. flexneri* and >16 μg/mL) for *S. sonnei*) was found in 45 isolates. This included 40 isolates of *S. flexneri*, 4 of *S. boydii* and one isolate of *S. sonnei.*The plasmid mediated resistance to azithromycin by *mph* A gene was detected in 19 out of 40 isolates of *S. flexneri.* The first report of DSA at our institute was in 2001 (n = 1, 2.5%) which gradually increased over 2006–10 (n = 4,4.1%) and finally to 35.1% (n = 40) in 2011–15. Amongst 45 DSA isolates, 35 isolates of *S. flexneri* and all DSA isolates of *S. boydii* and *S. sonnei* (n = 5) were isolated in 2015.

Table 1
Primer sequences for PCR of antimicrobial resistance genes used in the present study.

Primer	Sequence
bla OXA	OXA-F 5′-ATGAAAAACACAATACATATCAACTTCGC-3′ OXA-R 5′-GTGTGTTTAGAATGGTGATCGCATT-3′
bla TEM	TEM-F 5′-ATGAGTATTCAACATTTCCCG-3′ TEM-R 5′-ACCAATGCTTAATCAGTGAG-3′
bla SHV	SHV-F 5′-AGCCGCTTGAGCAAATTAAA-3′ SHV-R 5′-CGCTGTTATCGCTCATGGTA-3′
bla CTXM-15	CTX-M-15-F 5′-CCAGAATCAGCGGCGCACGA-3′ CTX-M-15-R 5′-GCGCTTTGCGATGTGCAGCA-3′
bla CMY-2	CMY-2-F 5′-AACACGGTGCAAATCAAACA-3′ CMY-2-R 5′-CCGATCCTAGCTCAAACAGC-3′
mphA	mphA-F 5′-GTGAGGAGGAGCTTCGCG AG-3′ mphA-R 5′-TGCCGCAGGACTCGGAGG TC-3′

4. Discussion

A change in the epidemiology of *Shigella* species and shift in the prevalent serogroups was observed in our study. Prior to 2003 the predominant *Shigella* species was *S. flexneri*. However, in 2003 ciprofloxacin resistant *S. dysenteriae* 1 caused an outbreak emerging as the predominant serogroup and in 2004, a switch back with resurgence of *S. flexneri* was seen showing a see-saw pattern in the prevalence rates.

Expansion in the prevalence of *S. sonnei* has been observed world over, probably, due to improved global water quality, immunization conferred by *Plesiomonas shigelloides* in contaminated water, survival of *S. sonnei* in free living amoeba like *Acanthamoeba* or due to its ability to maintain a greater array of antimicrobial resistance genes [8]. We too, observed a rise in the prevalence of *S. sonnei* and similar trends were observed with *S. boydii*. Outbreaks due to *S. flexneri* and *S. sonnei* have been reported from across India viz. West Bengal (2007), Kerala (2009) and Maharashtra (2010) [1]. An increase in the prevalence of *S. sonnei* with a fall in *S. flexneri* has been documented in some studies [9]. In a recent study from our center, the reappearance of *S. dysenteriae* has been reported in 2019, although *S. flexneri* still remains the most common sero-group followed by NAG [10].

We did not detect NAG till the year 2003–2004 but their occurrence rose to 15.2% in 2011–2015. Other Southern and East Indian studies have reported a 5%–13% prevalence of NAG *Shigella* isolates [9,11]. In a study by Anandan et al., the isolation rate of NAG was found to be 8.5% in 2014 with decrease to 1.2% in 2015 [11]. Isolation of uncommon, provisional or atypical strains or novel subserotypes of *Shigella* spp. is reported from across the world, including Bangladesh, Indonesia, Argentina, China, Japan and India [12,13]. The emergence of one such multidrug resistant variant of *S. flexneri* serotype X was reported from China as an epidemic clone superseding the serotype 2a [14]. Despite the epidemiological importance, exclusion of these serotypes from the current serological scheme exemplifies the need for updated serotyping along with systematic and continuous surveillance of circulating *Shigella* species/serotypes [1].

Before the 1990's, the preferred drug for treatment of shigellosis was nalidixic acid. However, subsequently, ciprofloxacin was recommended in view of high-level resistance to nalidixic acid, ampicillin, chloramphenicol and co-trimoxazole. Till 2000, ciprofloxacin resistance in *S. flexneri* was infrequent and sporadic across the world including our center. In 2001, ciprofloxacin resistant *S. dysenteriae* serotype 1 caused an epidemic in north-east India, and in 2003, fluoroquinolone resistance was reported at Kolkata, India in *S. flexneri* type 2a [15]. There are also reports of quinolone resistance in *S. sonnei* and *S. boydii* world over and from India [16,17]. At our center, ciprofloxacin resistance in *S. boydii* was limited to a single isolate in 2005 increasing to 4 cases in 2015.

Current recommendations by WHO support the use of ceftriaxone and azithromycin for the management of fluoroquinolone-resistant shigellae [4]. However, over reliance on these drugs should be avoided. A previous study from our center revealed that 15.1% isolates of *S. flexneri* collected from 2000 through 2009 were resistant to at least one third-generation cephalosporin [18]. Statistically significant rise in the proportion of cephalosporin resistant shigellae has also been noted in other studies from India [19]. In 2001, the first ceftriaxone resistant isolate was

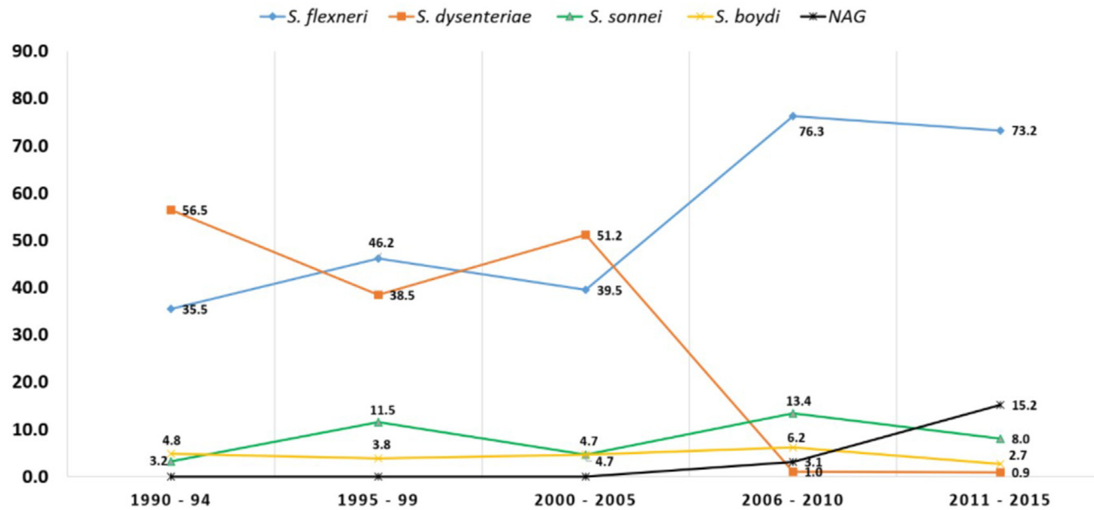

Fig. 2. Trends in species distribution of *Shigella* spp from 1990 to 2015.

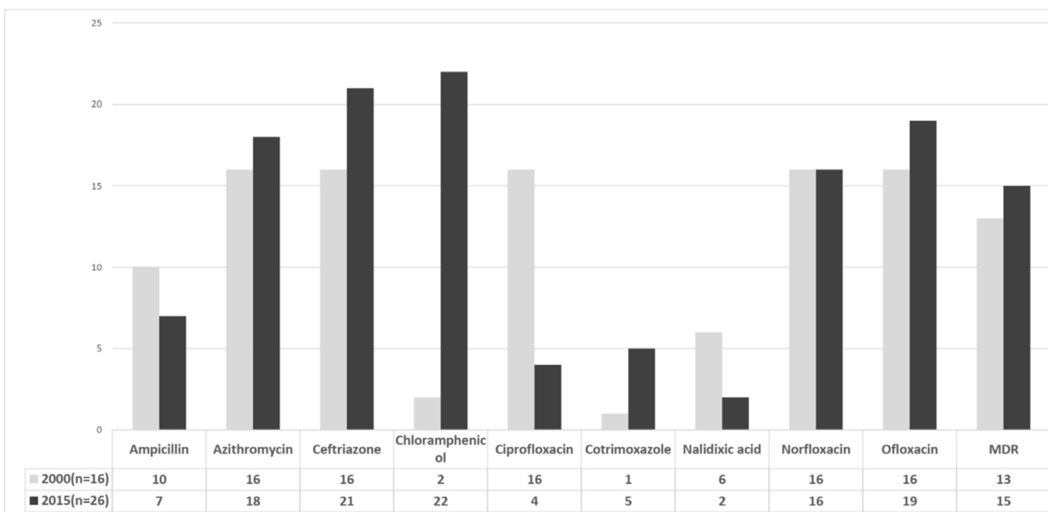

Fig. 3. Comparison of antimicrobial susceptibility results of *Shigella* species in 2000 and 2015.

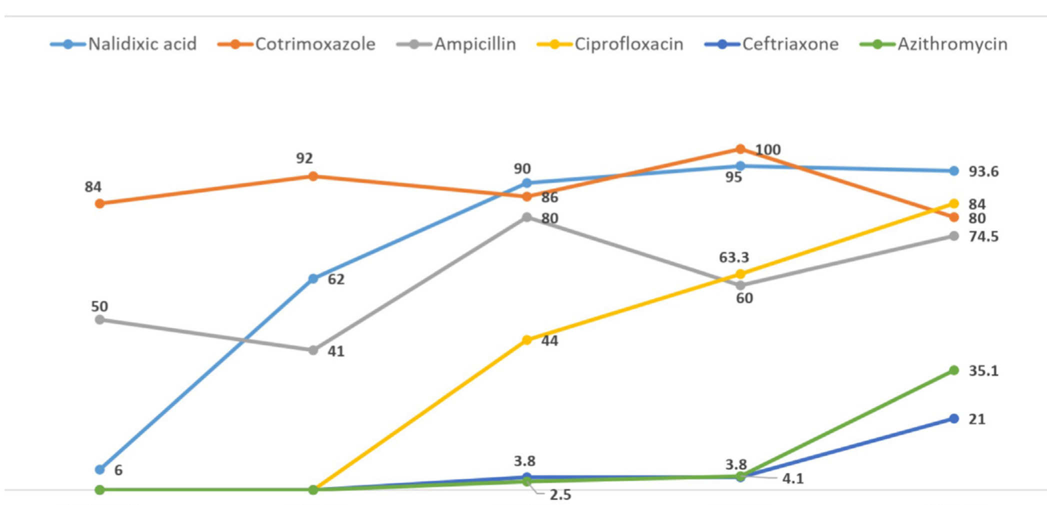

Fig. 4. Trends in antimicrobial resistance from 1990 to 2015.

obtained at our center followed by a worrisome increase in resistant *S. flexneri*. Although the MIC values for *S. dysenteriae* remained < 1 mg/L for ceftriaxone the MIC_{90} for cefepime have risen to 4 mg/L. In ceftriaxone resistant isolates the percentage prevalence of *bla*TEM (100%) was highest among the ESBL genes which was similar to a previous study from our center [18]. However, in the present study a comparative decrease in *bla*CTXM-15 (from 50% to 37%) and *bla*CMY-2 (from 45.6% to 35%) with increase in *bla*OXA (from 40% to 52%) was seen. Isolates from Bay of Bengal islands, showed a 2% prevalence of the *bla*SHV gene contrary to none seen in the present and in a previous study from our center [20]. *Shigella* producing CTX-M ESBL have been previously reported from various countries including Shanghai (CTX-M-14 and-15) [21], Korea (CTX-M-14) [22], Argentina (CTX-M-2) [23] Vietnam (CTX-M-15 and -24) [24] and Turkey (CTX-M-3) [25]. In India, CTX-M-3 has been reported previously in an isolate of *S. flexneri* from Andaman and Nicobar Islands [26]. The high prevalence of CTX-M-15 in isolates of the present study is concerning as it has potential of mobilization during oro-faecal passage and may have serious connotations on the further spread of resistance.

In our study, the emergence of DSA amongst shigellae in 2001 followed by the rise to 35.1% in 2015 is disconcerting. The proportion of DSA *Shigella* also showed an increase from none in 2000–2002 to 50% in 2006–2009, from Andaman and Nicobar Islands, India [19]. We detected plasmid mediated resistance to azithromycin mediated by *mph* A gene in 47.5% *S. flexneri* with DSA. In a study from United States of America, 55% of the isolates with DSA had *mph*A mutations alone while 43% contained mutations in both *mph*A and *erm*B gene [27]. Patients infected with *S. sonnei* containing *mph*A and *erm*B genes with azithromycin MIC >16 μg/mL are reported to have treatment failure, suggesting the importance of clinical breakpoints to indicate the use of alternative therapeutic options [28,29].

Limitations of the study are that we were unable to check the molecular determinants of resistance of many antibiotics. Since the local epidemiology of our region has previously indicated the high prevalence of the beta-lactamase genes we focused on these and DSA. Future studies looking into other molecular mechanisms are needed to elucidate the complete antimicrobial resistance profile. We were also unable to recover 49.1% of the isolates from our culture collection, primarily from the years 2000–2005 and this may also have impacted the findings. Nonetheless, the emergence of antibiotic resistant *Shigella* calls for stringent and systematic monitoring and periodic updates of countrywide and local antibiogram patterns and this study provides useful insight into these trends over a long duration of surveillance.

5. Conclusion

The scientific community must focus efforts to identify the drugs most suitable for empirical therapy according to these antibiograms. To decrease the morbidity and mortality associated with diarrhea in our country we not only need newer antimicrobials, cost effective vaccines but also public health measures such as safe drinking water and sanitations [30]. A combination of all these efforts may help in limiting the global expansion of antimicrobial resistant shigellae and thwart this eminent threat.

Author contributions

The concept of study was formulated by NT. Data collection was carried out by A.Me, AK, A.Mi. Laboratory work was carried out by KZ and PG. Data analysis and initial draft preparation was performed by KZ and SS. NT provided critical inputs during manuscript preparation and data analysis. NT is designated as 'guarantor' to take responsibility for the integrity of the work as a whole from inception to published article.

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Declaration of competing interest

The authors have no conflict of interest to declare.

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